Bank Customer Churn Analysis Using Python

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An Exploratory and Statistical Investigation of Customer Attrition Patterns, Banking Behaviour, and Retention Opportunities

1. Project Overview

This project analyzes bank customer data to explore customer churn patterns and identify characteristics associated with customers leaving the bank. Using Python, the analysis examines customer demographics, geographical differences, account balances, and membership activity to understand how churn varies across customer groups.

The project applies data inspection, data cleaning and transformation, statistical analysis, and data visualization to convert raw customer records into meaningful findings and business recommendations.

The main objective is to identify customer segments that may require further investigation and support evidence-based customer retention decisions. The project focuses on exploratory data analysis and statistical investigation; machine learning is not included in the completed analysis.

2. Data Source

- **Dataset: Bank Customer Churn**
- **Source: Maven Analytics Data Playground**
- **Domain: Banking and Customer Analytics**
- **File Format: CSV**
- **Dataset Size: 10,000 customer records and 13 columns**
- **Analysis Environment: Google Colab**
- **Programming Language: Python**

The dataset contains customer demographic, financial, and banking-behaviour information. The Exited column is the primary outcome variable, where 1 indicates that a customer exited the bank and 0 indicates that the customer stayed.

3. Problem Statement

Customer churn is an important concern in the banking industry because losing customers can affect long-term customer relationships and business

performance. Customer databases contain demographic, financial, and account-activity information, but these attributes require systematic analysis to identify patterns associated with customer exits.

This project investigates the following question:

Which customer characteristics and banking behaviours are associated with higher customer churn, and what retention opportunities can be identified from the available data?

The analysis identifies statistical patterns and associations. It does not establish that any particular characteristic causes customers to leave.

4. Project Objectives

1. **Inspect the dataset structure, columns, and data types.**
2. **Prepare the dataset for analysis through data cleaning and transformation.**
3. **Examine the distribution of customer churn.**
4. **Explore customer age distributions and potential outliers.**
5. **Compare churn rates across geographical regions.**
6. **Investigate the association between membership activity and customer churn.**
7. **Compare age and account-balance distributions between customers who stayed and exited.**
8. **Examine churn patterns across combinations of age groups, geography, and membership activity.**
9. **Develop visualizations and interpret their business implications.**
10. **Summarize key findings and recommend areas for further customer-retention investigation.**

5. Dataset Attributes

The dataset contains the following 13 columns:

Attribute	Description
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Customer credit score

Geography	**Customer's country**
Gender	**Customer gender**
Age	**Customer age**
Tenure	**Number of years with the bank**
Balance	**Customer account balance**
NumOfProducts	**Number of banking products held**
HasCrCard	**Credit card ownership indicator**
IsActiveMember	**Active membership indicator**
EstimatedSalary	**Estimated customer salary**
Exited	**Customer churn indicator**

The CustomerId and Surname columns primarily identify customers and are not direct measures of customer behaviour. The Exited column is used to distinguish customers who stayed from those who left.

6. Tools and Technologies

- **Python: Main programming language for the analysis.**
- **Pandas: Dataset loading, data manipulation, grouping, and summary tables.**
- **NumPy: Numerical calculations and data analysis.**
- **Matplotlib: Chart creation and visualization.**
- **Seaborn: Statistical plots, including count plots, histograms, box plots, bar charts, and heatmaps.**
- **Google Colab: Notebook development and execution.**
- **GitHub: Source code storage, version control, and portfolio presentation.**

7. Project Workflow

Stage 1: Project Setup and Initial EDA

The CSV dataset was loaded directly from the GitHub repository into a Pandas DataFrame. The initial exploration included examining the first records, dataset dimensions, column names, and data types to understand the structure of the data.

Stage 2: Data Cleaning and Transformation

The project workflow includes checking data quality, identifying duplicates and missing values, reviewing data types, and preparing the dataset for analysis. Data transformations and derived categories are used where required to support the visual analysis.

Data-quality decisions should preserve valid customer records and legitimate zero account balances. Unusual values should be investigated rather than removed automatically.

Stage 3: Statistical Analysis and Visualization

The project includes nine visualizations organized into three categories. Each visualization is accompanied by an interpretation explaining what the chart shows, the features used, the observed findings, and the potential business implications.

A. Univariate Analysis — 3 Visualizations

1. **Customer Churn Distribution — Count Plot: Compares the number of customers who stayed and exited.**
2. **Customer Age Distribution — Histogram: Examines the distribution of customer ages and identifies common age ranges.**
3. **Customer Age Distribution — Box Plot: Summarizes the median, spread, and potential age outliers.**

B. Bivariate Analysis — 4 Visualizations

4. **Customer Churn Rate by Geography — Bar Chart: Compares churn percentages across France, Germany, and Spain.**
5. **Customer Churn Rate by Active Membership — Bar Chart: Compares churn rates among active and inactive members.**
6. **Customer Age by Churn Status — Box Plot: Examines differences in age distributions between customers who stayed and exited.**
7. **Account Balance by Churn Status — Box Plot: Compares account balances between the two customer groups.**

C. Multivariate Analysis — 2 Visualizations

8. **Churn Rate by Age Group and Geography — Heatmap: Examines how churn rates vary across age groups and countries.**
9. **Churn Rate by Geography and Active Membership — Grouped Bar Chart: Compares churn rates across countries while distinguishing active and inactive members.**

Stage 4: Documentation, Insights, and Presentation

The final stage summarizes the results in plain English, identifies important patterns, distinguishes descriptive and diagnostic analysis from future predictive modelling, and proposes business recommendations based on the observed findings.

8. Key Findings

8.1 Overall Customer Churn

The dataset contains 10,000 customer records.

- **Customers who stayed: 7,963 (79.63%)**
- **Customers who exited: 2,037 (20.37%)**

The overall churn rate is 20.37%, meaning approximately one in five customers in the dataset exited the bank.

8.2 Churn Rate by Geography

Geography	Churn Rate
France	16.15%
Germany	32.44%
Spain	16.67%

Germany has the highest overall churn rate among the three countries, at 32.44%. Its rate is substantially higher than those of France and Spain, suggesting that retention patterns differ by geography.

8.3 Churn Rate by Membership Activity

Membership Status	Churn Rate
Active	14.27%
Inactive	26.85%

Inactive members have a higher churn rate than active members. The difference is 12.58 percentage points, indicating an association between membership activity and customer retention.

8.4 Customer Age by Churn Status

Customer Status	Average Age	Median Age

Exited	**44.84 years**	**45 years**
Stayed	**37.41 years**	**36 years**

Customers who exited were generally older than those who stayed. The age-group analysis also showed elevated churn rates among customers aged 41–60, particularly in Germany.

8.5 Account Balance by Churn Status

Customer Status	Average Balance	Median Balance
Exited	**91,108.54**	**109,349.29**
Stayed	**72,745.30**	**92,072.68**

Exited customers had higher average and median account balances than customers who stayed. This pattern warrants further investigation but does not establish that account balance causes churn.

8.6 Churn Rate by Age Group and Geography

The heatmap showed the highest observed churn rate among customers aged 51–60 in Germany, at 69.55%.

Other elevated rates included:

- **Customers aged 51–60 in France: 52.59%.**
- **Customers aged 41–50 in Germany: 48.82%.**
- **Customers aged 51–60 in Spain: 45.99%.**

These results identify geographical and age-group combinations that warrant further investigation.

8.7 Churn Rate by Geography and Membership Activity

Inactive members had higher churn rates than active members in all three countries.

Geography	Active Members	Inactive Members
France	**11.50%**	**21.13%**
Germany	**23.72%**	**41.08%**
Spain	**10.75%**	**23.35%**

Germany had the highest churn rates for both membership groups. The difference between active and inactive members was also substantial in Germany, highlighting a potential area for focused customer-retention investigation.

9. Types of Analysis

Descriptive Analysis

Descriptive analysis summarizes the customer dataset using counts, percentages, summary statistics, and visualizations. In this project, it describes the overall churn distribution, age distribution, and account-balance characteristics of customers.

Diagnostic Analysis

Diagnostic analysis explores differences in churn across customer groups. The project compares churn rates by geography, membership activity, age group, and combinations of age group and geography.

These comparisons identify associations and potential areas for further investigation; they do not prove why customers leave the bank.

Predictive Analysis

Predictive modelling was not included in the completed project. As a future extension, a classification model could estimate the likelihood of customer churn and be evaluated using precision, recall, F1-score, accuracy, and a confusion matrix.

Prescriptive Analysis

Prescriptive analysis considers potential actions based on the observed results. Recommendations include investigating high-churn geographical segments, reviewing engagement among inactive members, and examining the needs of customers in age groups with higher observed churn.

These are proposed actions and should be validated using further analysis and business evidence.

10. Recommendations and Decision Support

1. **Investigate Retention Challenges in Germany: Examine customer experiences, product suitability, and service expectations in Germany, where the overall churn rate is highest.**
2. **Improve Customer Engagement: Investigate why inactive members have higher churn rates and evaluate targeted engagement initiatives.**

3. **Review Age-Related Patterns: Examine the needs of customers aged 41–60, especially those in Germany, to identify possible retention opportunities.**
4. **Investigate Higher-Balance Customers: Explore whether customers with higher balances experience different service needs or retention challenges.**
5. **Monitor Churn Rates Regularly: Track overall and segment-level churn rates to identify changes and evaluate retention initiatives over time.**

These recommendations are based on observed patterns rather than proven causes of churn.

11. Limitations

- **The analysis is based on a single customer-level dataset and may not represent current banking behaviour.**
- **The available variables do not establish the reasons customers exited.**
- **The analysis identifies associations rather than causal relationships.**
- **Some age and geography combinations may contain fewer customers than others; group sizes should be reviewed when prioritizing retention efforts.**
- **No predictive machine learning model was built or evaluated as part of this project.**

12. Future Enhancements

Potential future improvements include:

1. **Using a larger or more recent banking dataset to investigate additional customer-retention patterns.**
2. **Adding transaction or customer-service interaction data for deeper analysis.**
3. **Conducting statistical significance tests for selected group comparisons.**
4. **Automating data-quality checks and analytical reporting using Python scripts.**
5. **Developing interactive dashboards using Power BI or Tableau.**
6. **Building and evaluating a machine learning model to estimate customer churn risk.**
7. **Monitoring churn trends over time if dated customer records become available.**

These enhancements are optional extensions and are not presented as completed project tasks.

13. Conclusion

The Bank Customer Churn Analysis project demonstrates the use of Python, statistical analysis, and visualization to explore customer attrition patterns. The results establish an overall churn rate of 20.37% and identify higher churn rates in Germany, among inactive members, and among certain age groups. Exited customers also had higher average and median account balances than customers who stayed.

The findings provide a starting point for further investigation into customer retention and demonstrate practical skills in data inspection, exploratory analysis, statistical interpretation, visualization, and business-focused reporting.

The project emphasizes the importance of interpreting results carefully, distinguishing associations from causes, and using evidence to guide future customer-retention analysis.